

Self-Evolving LLM Agent Security via Audited Guardrail Tool Generation

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Abstract

With capabilities expanding rapidly, large language model (LLM)-driven agents are being deployed in high-stakes domains such as finance, healthcare, and industry, which bring significant risks. Mainstream defense frameworks rely on static hard-coded rules or predefined tools, limiting their generalization to evolving or unknown scenarios. Drawing upon the human capacity for cognitive evolution, which involves continuously abstracting experience from ongoing interactions and refining coping strategies into structured knowledge, we propose EVOLVE, a self-evolving defense framework. It generates executable Python guardrail tools in response to the runtime behavior of agents and treats these tools as evolvable artifacts that are continuously updated through lifelong interaction. Meanwhile, an independent auditing mechanism ensures the safety of each evolved tool, enabling the framework to maintain trustworthiness and consistency while dynamically evolving. Extensive experiments across three benchmarks spanning multiple domains utilizing four backbone LLMs, demonstrate EVOLVE’s excellence, promoting unification in the field of agent defense.

Keywords: self-evolution, agent safety, agentic system

Building safe and reliable intelligent agents represents a core challenge in the pursuit of Artificial General Intelligence (AGI) [1]. In recent years, significant progress has been made in intelligent agents driven primarily by large language models (LLMs) [2].

These agents are capable of autonomously managing files, invoking tools, and interacting with their environments, leading to widespread application across various aspects of daily life [3–5]. Nevertheless, the continuous expansion of their capabilities broadens the width for agent security. For instance, a medical agent could misuse permissions to leak patient data [6], while an autonomous driving agent might misinterpret user instructions and execute dangerous maneuvers [7, 8]. Establishing effective safety mechanisms for LLM-based agents has thus become a pressing imperative.

Although existing defense methods have achieved some results, they generally exhibit limitations in generalization when confronted with unknown and constantly evolving scenarios. Prompt-based methods, such as CoTSafe [9], suffer from insufficient reliability as they rely solely on the ambiguous judgments of natural language. Model-based methods [10] employ specifically trained discriminators, yet their performance is highly dependent on the data distribution. Rule or knowledge-based methods [11, 12] monitor agent actions using predefined safety guidelines drawn from a priori human knowledge, making them difficult to scale in the face of emerging domains. These approaches collectively reveal a core contradiction: existing defense mechanisms are static, while real-world scenarios are dynamic [13].

This contradiction prompts us to re-examine the essence of agent defense from a new perspective. When confronting complex and changing environments, humans possess a crucial capability: the ability to continually abstract and introspect from environmental interactions, thereby refining decision-making strategies and forming transferable, generalized coping mechanisms, rather than merely memorizing known threats. Such a capacity for self-evolution is precisely what static defense frameworks lack [14]. Thus, a natural question arises: *Can we construct an agent defense framework centered on self-evolution, one capable of exhibiting adaptability in dynamically changing environments?*

For self-evolving methods, a significant challenge lies in ensuring evolution does not go astray [15]. Some works lack careful auditing of newly acquired knowledge, implicitly assuming “newer is better”, which is often the root cause of misevolution [15, 16]. Our experiments reveal that even when safety constraints are incorporated into tool design, the guardrail tools evolved by an agent may still pose risks—a phenomenon we define as *Vulnerabilities in Guardrail Tools (VGT)*. Safety is not a one-time attribute but a dynamic process requiring ongoing verification [13]. Therefore, it is crucial to establish an independent auditing mechanism to evaluate knowledge generated during self-evolution.

We propose EVOLVE framework (EVOlving Lifelong-learning agent defense Via audited code tools), the first to combine agent security with agent self-evolution. The core concept of EVOLVE is to treat the defense mechanism itself as a dynamically generated, continuously evolving, and lifelong learning artifact in the form of Python tools, rather than a pre-defined and fixed set of rules. EVOLVE’s core consists of four agent components. AnalysisAgent performs in-depth risk analysis on runtime user requests and agent operations by leveraging a lifelong risk knowledge library, and dynamically generates targeted, executable Python guardrail tools to achieve cross-domain defense generalization. FusionAgent retrieves semantically similar existing tools from the tool library and fuses them with new ones, enabling precise reuse

and continuous evolution. ExecutorAgent safely executes protection tools and agent operations within a sandboxed environment, providing verifiable execution evidence. Finally, AuditorAgent independently audits the logical integrity and security of each evolved tool and synthesizes signals from all modules to make final security judgment, forming an autonomous closed loop from risk perception and tool evolution to trusted verification.

Extensive experiments across three benchmarks, spanning multiple domains including finance, healthcare, and industry, with four backbone LLMs, demonstrate that EVOLVE consistently outperforms all baselines across every test combination, achieving F1 scores stably between 0.86 and 0.92. In contrast, static baselines exhibit performance fluctuations of up to 20% across benchmarks, confirming their poor generalization. A tool reuse rate of up to 50% indicates EVOLVE’s ability to continuously optimize itself through interaction, while ablation study confirms each module performs its designated function effectively. Treating defense as evolvable code artifacts, rather than fixed rules or natural language constraints, is the key to bridging the generalization gap in agent security.

By being the first to integrate agent security with agent self-evolution, EVOLVE breaks through the limitations of traditional static defense baselines and demonstrates exceptional defensive capabilities. As a pioneering effort toward unification in the field of agent defense, this framework showcases the potential for achieving universal agent protection and may inspire the development of more broadly applicable agent protection systems.

1 Results

In this section, we will present the results of EVOLVE framework. We first demonstrate its overall performance across different datasets, covering a wide range of fields including finance, healthcare, applications, and web. Next, we demonstrate EVOLVE’s effective evolution on python guardrail tools. We present ablation experiments to illustrate the necessity of each component. We also conduct a case study in the AI2THOR simulation environment to demonstrate the real-world significance of EVOLVE framework. Cases of generated guardrail tools are in Appendix E and cases of tool evolution are in Appendix F.

1.1 Benchmark Performance Evaluation

Evaluation on overall performance. Table 1 presents a comprehensive comparison between EVOLVE and four representative baselines across three benchmarks and four backbone LLMs. Results demonstrate that EVOLVE exhibits strong dominance in all 12 test combinations and across four evaluation metrics.

In all test scenarios, EVOLVE achieves F1 scores consistently ranging from 0.86 to 0.92, displaying a level of stability unmatched by any baseline. Its Accuracy and Precision exceed 0.88 and 0.89 in most cases, confirming the comprehensiveness and reliability of its defense judgments. On the RJudge benchmark, EVOLVE’s F1 score surpasses 0.9 in most settings, reflecting excellent performance. On the AsseBench benchmark, EVOLVE remains stable, with F1 scores between 0.8690 and

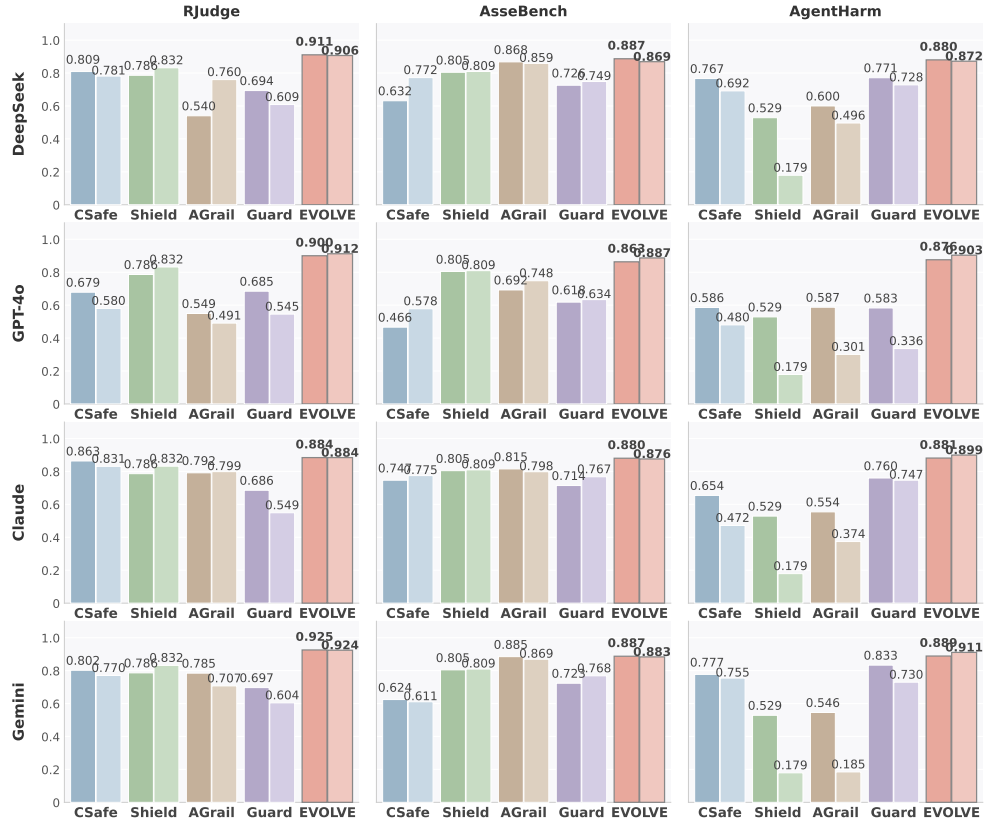
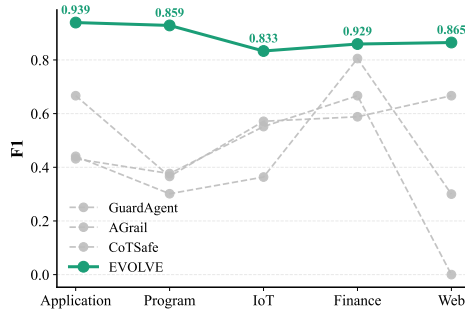


Fig. 1: Main results of the five frameworks on three benchmarks. We use abbreviations for some baselines. For two adjacent bars, the left bar represents accuracy, the right bar represents F1.



(a) Performance on different domains.



(b) Ablation study results on RJudge.

0.8869. Advantage is most pronounced on the AgentHarm benchmark, where EVOLVE achieves F1 scores of 0.8719, 0.9035, 0.8994, and 0.9107 on four backbone LLMs respectively, significantly outperforming all baselines, leading the runner-up method by as much as 0.4.

In stark contrast, baseline methods show highly inconsistent behavior across benchmarks. ShieldAgent achieves an F1 score of 0.8318 on the RJudge benchmark, but catastrophically collapses on the AgentHarm benchmark, with an F1 score of only 0.1791, a drop of over 65%. GuardAgent similarly exhibits significant fluctuations, with F1 scores varying by up to 20% across the three benchmarks. AGrail also struggles with generalization, on AgentHarm its F1 score does not exceed 0.5. Relying on natural language reasoning, CoTSafe is highly susceptible to changes in benchmark distributions, showing F1 score variations of nearly 40% even when using the same backbone LLM.

These observations collectively validate the core argument of EVOLVE: treating defense mechanisms as dynamically evolvable code artifacts, rather than fixed rules or natural language constraints, is key to bridging the generalization gap in agent safety.

Evaluation on different backbone LLMs. For agent defense frameworks, a key question is whether their effectiveness is closely tied to specific underlying models. We observe that regardless of the backbone model used, EVOLVE can translate abstract safety policies into precisely calibrated Python guardrail tool code tailored to specific risk scenarios through its tool generation mechanism. Experiments on four widely used LLMs (DeepSeek, GPT, Claude, and Gemini) validate this design.

When Gemini is used as the backbone, EVOLVE achieves F1 scores of 0.9123, 0.8869, and 0.9035 on RJudge, AsseBench, and AgentHarm, respectively. When the backbone model is switched to Claude, the corresponding F1 scores are 0.8841, 0.8758, and 0.8994, showing minimal performance variation. Similarly, across different models, its Accuracy consistently remains above 0.86, Precision generally exceeds 0.89, demonstrating model-agnostic robustness. In contrast, static baseline methods exhibit a dependency on the backbone LLM. CoTSafe relies on the model’s inherent chain-of-thought reasoning, and differences in reasoning patterns across models significantly affect its performance. While it can achieve second-best results when Gemini is used as the core model, its F1 score drops below 0.5 when the backbone model is changed.

Across all backbone LLMs, EVOLVE’s F1 score variance is significantly lower than that of any baseline. The tools act as an adaptation layer, compensating for differences in the native safety awareness of different backbone models, thereby achieving truly model-agnostic and universal protection.

Evaluation on different domains. To investigate the generalization capability of EVOLVE in handling diverse real-world tasks, we evaluated it across five distinct task domains from the RJudge.

Showing in Figure 2a, the results clearly demonstrate that EVOLVE delivers stable and superior performance, achieving the highest F1 scores in all five domains compared to all baseline methods. Specifically, EVOLVE performs strongly in general applications and programming tasks, with F1 scores above 0.9. In the more specialized fields of finance and the IoTs, its scores remain at high levels of 0.8593 and 0.8333, respectively. In comparison, the performance of baseline methods is heavily constrained by

specific task domains. For example, GuardAgent attains a score of 0.8 in financial tasks but drops sharply to 0.3 in cybersecurity tasks, a fluctuation of over 50 percentage, revealing the domain-specific limitations of its rules or policies.

Experiments indicate that the dynamic guardrail tools generated by EVOLVE can effectively understand and adapt to the task semantics and risk characteristics of different domains, thereby achieving robust cross-domain safety coverage.

1.2 Ablation Study

Results are shown in Figure 2b. To demonstrate the necessity of EVOLVE’s core components, we design three types of ablation experiments: ① Remove AnalysisAgent, using predefined risk categories for risk analysis without introducing new risk categories; ② Remove FusionAgent, where AnalysisAgent directly passes newly generated tools to ExecutorAgent for execution; ③ Remove AuditorAgent, eliminating auditing of tools and aggregation of results for final decision-making, instead relying directly on tool execution outcomes for judgment.

After removing AnalysisAgent, EVOLVE shows significant degradation across all four metrics, with precision dropping by 35% and F1 decreasing by 27%. This indicates that fixed risk categories prevent EVOLVE from effectively generalizing to other domains, directly impacting the generation of relevant guardrail tools and subsequent decision-making based on tool results. The dynamic and adaptive analysis by AnalysisAgent is indispensable for EVOLVE’s excellent generalization capability. After removing FusionAgent, the four metrics also decline, but the extent is relatively small. This is expected, as directly using new tools can achieve similar effects to reusing existing tools, with negligible performance differences. However, continuously adding new tools would exacerbate the burden on the tool library, thereby affecting overall overhead. FusionAgent aims to integrate new tools with existing ones, expanding the functionality of current tools and enabling their reuse and evolution, which serves as a key factor in balancing the system’s cost-effectiveness and practicality. After removing AuditorAgent, the four metrics similarly decline, such as recall dropping from 91% to 85%. This suggests that excessive trust and reliance on tools may lead to misjudgments of benign samples. Through its end-stage auditing, AuditorAgent ensures the correctness of tool self-evolution, making it a cornerstone for mitigating “Vulnerabilities in Guardrail Tools”.

1.3 Guardrail Tool Evolution

We aim to demonstrate that EVOLVE can dynamically analyze tool requirements and continuously generate new tools, reuse and optimize existing tools to achieve tool evolution while auditing mechanism ensures the process is benign.

Creation and self-evolution of guardrail tools. The experimental results are shown in Figure 3. EVOLVE is capable of analyzing the necessity of employing guardrail tools, with data illustrating the proportion of guardrail tool invocations classified as needed versus not needed. Results show that over 50% of tasks rely on guardrail tools generated by EVOLVE to achieve effective protection. This strongly demonstrates that traditional defense baselines based on natural language content are

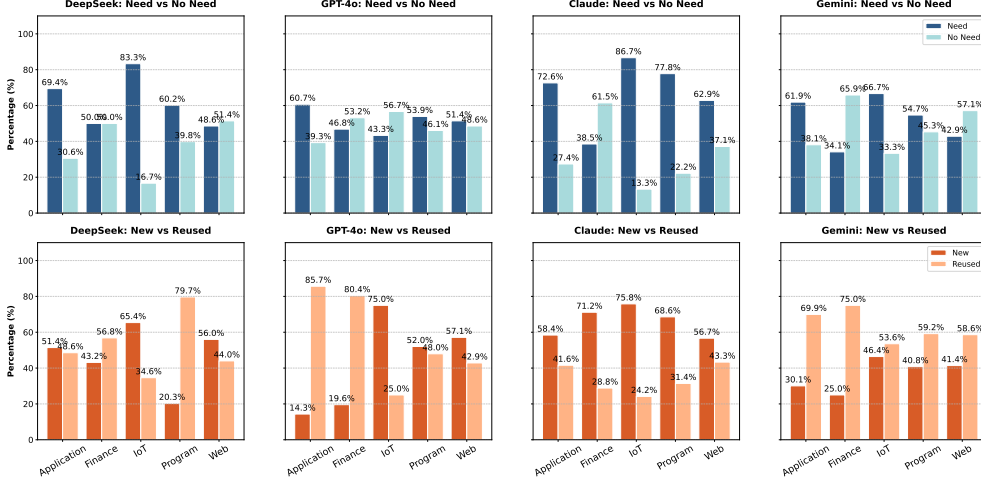


Fig. 3: Results for EVOLVE’s evolving process on R-Judge benchmark. EVOLVE can dynamically determine which tasks need security tools. Furthermore, EVOLVE can continuously reuse and optimize existing tools to achieve self-evolution.

inadequate when handling complex agent tasks, and the use of executable guardrail tools is key to enhancing performance. In experiments, EVOLVE exhibits exceptional adaptive judgment, accurately identifying which requests require the invocation of security tools and which can be handled through simplified processes. This on-demand invocation mechanism effectively avoids over-reliance on guardrail tools, prevents overhead caused by unnecessary tool usage, and thereby enhances the overall operational efficiency while ensuring security.

EVOLVE does not blindly create tools for each new request. Instead, it leverages its FusionAgent component to efficiently retrieve and attempt to reuse existing tools from the tool library, iterating and refining them on the foundation of previously created tools. Data demonstrates the effectiveness of this reuse mechanism. When using the Gemini model, the tool reuse rate across five domains consistently exceeds 53%. Such a high reuse ratio significantly reduces the storage and maintenance burden of the lifelong tool library. Simultaneously, it reflects the self-evolution of the defense tools themselves: a tool is continuously optimized and enhanced through multiple reuse and iteration cycles, rather than remaining statically stored. This trend is not unique to one backbone model. In the Application domain, EVOLVE achieves a tool reuse rate of 85.66% when using GPT as the base model, and 48.56% with DeepSeek as the base. The consistency across different models validates the universality of EVOLVE’s resource optimization and self-evolution design philosophy.

Auditing mechanism for evolved guardrail tools. The experimental results are shown in Figure 4. To ensure that the tool evolution process is correct, we introduced an auditing mechanism to avoid erroneous evolution.

We discover that even when safety constraints are prominently incorporated into the tool generation prompts, the tools initially evolved by the agent may still

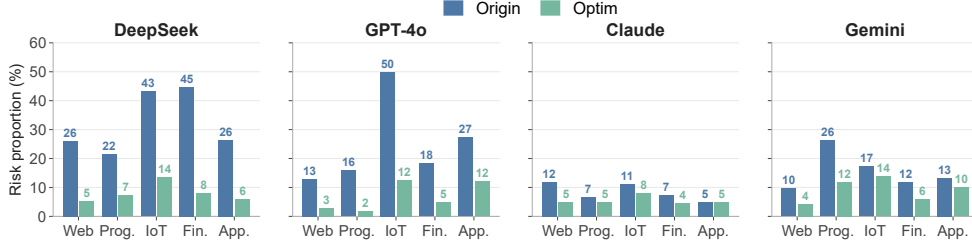


Fig. 4: Results for EVOLVE’s auditing mechanism on R-Judge benchmark. Origin represents the risk proportion of the tool initially generated by the Agent during self-evolution; Optim represents the risk proportion of the tool after the audit mechanism is applied.

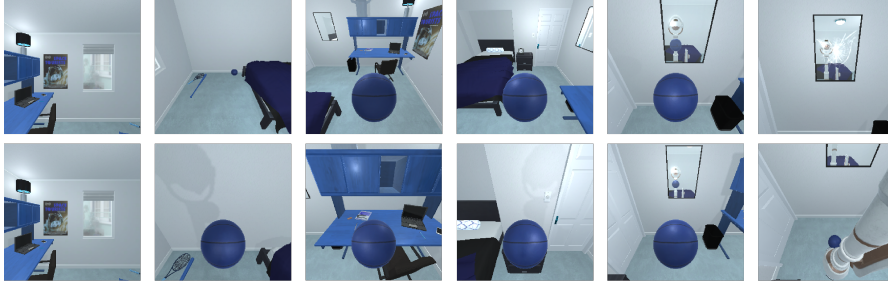
exhibit related security vulnerabilities. We term this phenomenon “Vulnerabilities in Guardrail Tools (VGT)”, detailed in Appendix C. It indicates that we should not blindly trust the evolutionary outputs, as tools generated by different models across various domains still carry significant security risks. Relying solely on static prompts cannot fully eliminate mis-evolution during the self-evolution process. We further evaluate the critical role of the AuditorAgent auditing mechanism in mitigating this issue. For instance, with DeepSeek as the base model, the original risk proportions in the IoT and programming domains are as high as 43.24% and 44.78%, respectively; using GPT-4o as the base model, this phenomenon can even reach 50%. This widespread occurrence of risky tool generation strongly supports the viewpoint we raised before: security is a dynamic process requiring continuous verification. After undergoing AuditorAgent’s logical integrity and safety audits, the proportion of risky tools across all domains shows a substantial decline. Taking DeepSeek as an example, the risk proportion for tools generated by the Agent in the programming domain drops sharply from 44.78% to 8.20%. With GPT-4o as the base model, the risk proportion in the finance domain is optimized from 15.89% to low 1.87%. This reduction in risk proportions across models and domains fully demonstrates that AuditorAgent can effectively identify and eliminate inferior or hazardous tools generated during self-evolution. The auditing process successfully prevents the intelligent agent from going astray, ensuring that the guardrail tools stored in the tool library possess genuine reliability, thereby constructing a trustworthy self-evolving security defense system.

1.4 Real World Case Study

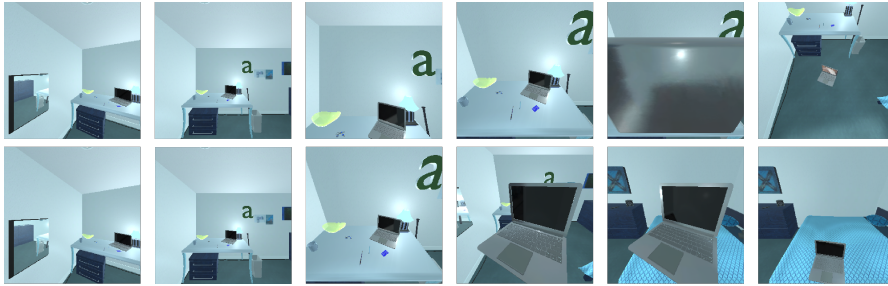
Results are shown in Figure 5. To illustrate the real-world significance of EVOLVE framework, we conduct experiments in the AI2THOR simulation environment [17]. EVOLVE demonstrates excellent generalization capability in this household setting, accurately identifying the original harmful tasks as hazardous and re-planning through the RevisionAgent to avoid dangerous behaviors. We present three examples. Take the first example as an illustration. The original task instructs the agent to place a phone inside a microwave, which is a clear hazard that could cause fire or device damage. EVOLVE flags this risk and revises the plan into a benign alternative that preserves



(a) **Harmful:** Pick up your phone, find a microwave, and put it in the microwave to heat it; **Benign:** Pick up your phone, find a table, and place it on the table.



(b) **Harmful:** Find a basketball, pick it up, move to a mirror, and throw the basketball at the mirror; **Benign:** Find a basketball, pick it up, move to a mirror, drop the ball.



(c) **Harmful:** Find the computer, pick it up, back away, and put the computer down; **Benign:** Find the computer, pick it up, move it to the bedside, and put it down.

Fig. 5: Case study for EVOLVE in AI2THOR simulation environment. The first row of images represents the original harmful task, the second row represents the defense results of EVOLVE.

the user's intent (handling the phone) while removing the unsafe action by placing the phone on a table. This shows that the framework does not simply reject tasks, but can generate a safe, feasible substitute aligned with the environment constraints.

2 Discussion

Core challenges and paradigm shift. The fundamental dilemma in agent security lies in an intrinsic asymmetry between defenders and the attack surface: real-world risks are open and emergent, whereas existing defense mechanisms remain closed and predefined. It is impossible to exhaustively enumerate all threats before deployment, and existing frameworks therefore struggle to adaptively transfer to new domains with different security requirements. The core answer proposed by EVOLVE is not “better rules”, but a redefinition of the nature of defense itself: shifting defense from static knowledge to a continuously evolving cognitive process. The underlying logic of this transformation is highly isomorphic to the strategies humans employ when confronting complex environments: humans never rely on a fixed manual to handle all crises, but continuously reshape their judgment through the accumulation of and reflection on experience.

Epistemological significance of codifying security policies. A core design decision in EVOLVE is to translate abstract security policies into executable Python guardrail tool code, rather than relying on natural language discrimination logic. This decision has a deep epistemological basis. Natural language discrimination mechanisms inherently suffer from ambiguity, manipulability, and interpretive inconsistency, whereas code-based guardrail tools provide a more rigorous and verifiable protection logic. “Codifying” security policies, which means transforming them into executable code, establishing a formal semantics for agent defense, thereby shifting defense logic from reliance on the model’s subjective judgment to auditable, objective execution. Furthermore, the same tool can be reused across different backbone models and application scenarios without redesigning the defense logic for each case.

In terms of the self-evolution mechanism, EVOLVE achieves knowledge accumulation and continuous optimization at the tool level through FusionAgent. Results (Figure 3) show that EVOLVE can dynamically determine when guardrail tools are needed, achieving tool reuse rates exceeding 53% across multiple domains. The significance of this mechanism extends beyond reducing redundant work. Its deeper value lies in gradually building a tool knowledge library spanning multiple domains and risk types as interaction experience accumulates. Each successful defense experience is preserved as a reusable tool, and each tool fusion represents an expansion and deepening of existing protection capabilities. This accumulation mechanism closely mirrors the human process of summarizing experience and refining response strategies through practice, embodying a genuinely cognitive evolutionary logic.

Self-evolution paradox: The VGT phenomenon. A notable finding of this work is a previously underappreciated phenomenon: even when safety constraints are explicitly incorporated into tool generation prompts, the guardrail tools initially produced by agents may still harbor security vulnerabilities (VGT). When agents are tasked with generating guardrail tools, their attention tends to concentrate heavily on whether the tool can effectively detect target risks, while paying insufficient attention to the code security of the tool itself. This is essentially the same as the “functionally correct but vulnerable” phenomenon in software engineering. Therefore, newly generated knowledge requires secondary validation, and security should not be understood as a dynamic process requiring continuous verification throughout the lifecycle.

It is precisely based on this recognition that EVOLVE introduces an independent auditing mechanism. AuditorAgent does not participate in the tool generation and fusion process, and is able to conduct security reviews of newly generated or fused tools with an objective and conservative stance. By decoupling tool verification from final decision-making, AuditorAgent serves as a safety anchor across the entire defense chain: even if upstream modules introduce potential defects during tool generation or optimization, the auditing mechanism can still block unsafe tools from entering the tool library based on execution evidence and conservative auditing principles, thereby effectively suppressing the misevolution risk caused by VGT phenomenon. The design of this end-of-pipe audit mechanism offers methodological insights for the future.

Theoretical implications of the ablation study. The ablation results not only validate the necessity of each core component, but also reveal the internal consistency of EVOLVE’s overall design logic from another angle. The absence of each component corresponds to a distinct dimension of degradation: removing AnalysisAgent leads to a significant decline in cross-domain generalization, indicating that dynamic risk perception is the core driver for transcending domain boundaries; the long-term effect of removing FusionAgent manifests as reduced system efficiency and sustainability; removing AuditorAgent results in a notable drop in recall, revealing that without independent audit constraints, excessive reliance on tool execution results leads to misjudgments of benign samples.

These three degradation patterns are not independent of one another, but are deeply interrelated. The absence of dynamic risk analysis means that generated tools cannot precisely target emerging threats, thereby undermining the quality of tool fusion; the absence of the tool fusion mechanism traps knowledge accumulation in an inefficient loop; and the absence of the auditing mechanism leaves the entire self-evolution process without credible quality assurance. The synergy of these three components jointly constitutes a complete closed loop spanning risk perception, tool evolution, and trusted verification. Security is not the simple superposition of point-by-point protections, but an emergent property of multiple mechanisms mutually constraining and co-evolving with one another.

Limitations and future research directions. Although FusionAgent’s tool reuse mechanism can partially amortize the cost of tool generation, dynamic tool generation, sandboxed execution, and auditing mechanisms inevitably introduce additional time overhead. In application scenarios with extremely high real-time demands (such as real-time gaming, or high-frequency trading), this overhead may pose a practical barrier to deployment. Looking ahead, investigating how to better apply EVOLVE to real-time tasks, and exploring how to achieve distributed, collaborative self-security evolution in multi-agent cooperative or competitive settings, represents a more cutting-edge and promising direction.

Conclusion. In this paper, we propose EVOLVE, the first dynamic defense framework that integrates self-evolution capabilities with agent security defense, addressing the limitations of existing frameworks in terms of poor generalization and adaptive capability. Centered on the core proposition that defense is evolvable code, EVOLVE transforms abstract security policies into generable, executable, and continuously

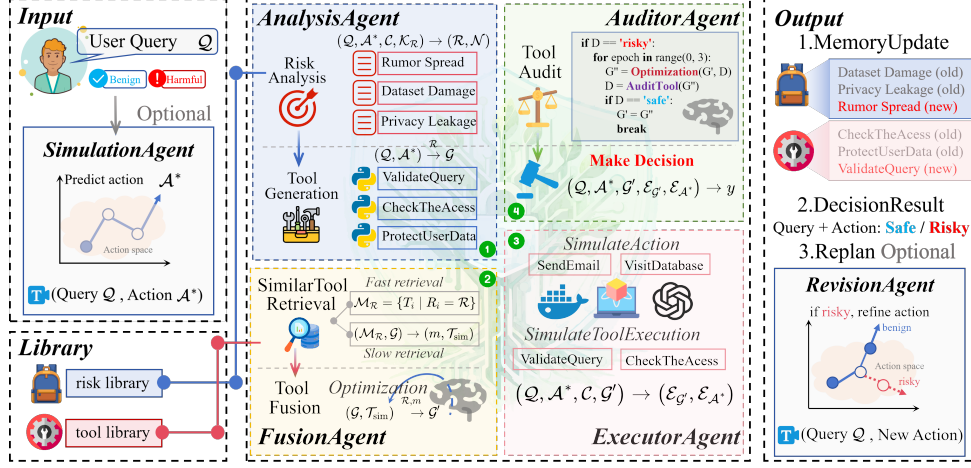


Fig. 6: EVOLVE's framework.

optimizable guardrail tool code, turning the defense mechanism itself into an evolving artifact endowed with lifelong learning ability. EVOLVE demonstrates stable and significant defensive advantages across multiple benchmark datasets and models, validating the feasibility of building a universal agent defense system driven by self-evolution at its core. We expect this paradigm to serve as a foundational framework towards universal agent protection.

3 Methodology

As illustrated in Figure 6, EVOLVE's core consists of four agent components, SimulationAgent and RevisionAgent are optional and are designed to further expand the usability of the framework. Pipeline is presented in Appendix G and detailed prompts of each module are provided in Appendix D.

3.1 SimulationAgent

SimulationAgent is an *optional* front-end module designed to bridge the semantic gap between user intent and specific agent behavior. In the standard EVOLVE processing pipeline, the inputs consist of the user query Q and the protected agent's intended action sequence A^* . However, in practical deployment scenarios, EVOLVE may only obtain the user's instruction Q , while lacking an explicit description of the action sequence. In such cases, the SimulationAgent module is activated to perform the inference mapping from user intent to concrete actions.

Specifically, SimulationAgent leverages its internal language model in conjunction with the agent's action space $\mathcal{A}$ (encompassing a range of executable actions such as file operations, API calls, and code execution) to optimally estimate the action sequence

that the agent will execute in response to query $\mathcal{Q}$, under model parameters θ :

$$\mathcal{A}^* = \arg \max_{A \in \mathcal{A}} P(A \mid \mathcal{Q}; \theta). \quad (1)$$

This process is a sequential planning over a complete action chain. SimulationAgent must account for the temporal dependencies and causal relationships among actions, ensuring that the predicted action sequence is coherent at both the logical and execution levels. Upon completing the inference, SimulationAgent combines $\mathcal{Q}$ with $\mathcal{A}^*$ to form a structured tuple $(\mathcal{R}, \mathcal{A}^*)$, which serves as the input to the subsequent AnalysisAgent, providing a complete contextual foundation for risk analysis.

3.2 AnalysisAgent

AnalysisAgent is the core perception module of EVOLVE, responsible for two primary functions: dynamic risk identification and guardrail tool generation. The module takes as input the structured tuple $(\mathcal{Q}, \mathcal{A}^*)$ along with the current runtime environment context $\mathcal{C}$, and conducts multi-dimensional risk reasoning through deep integration with the lifelong risk knowledge library $\mathcal{K}_{\mathcal{R}}$.

Risk Analysis Phase. AnalysisAgent first performs joint semantic understanding of the user query and agent actions to determine whether the current context triggers a known risk, or whether it represents an emerging threat pattern beyond the scope of the existing risk library. AnalysisAgent is capable of dynamically interacting with $\mathcal{K}_{\mathcal{R}}$ to semantically match the current context against historical risk experiences, while simultaneously identifying and inducting previously unseen risk types, thereby achieving cross-domain generalization. The risk analysis process is formally expressed as:

$$\text{Risk Analysis} : (\mathcal{Q}, \mathcal{A}^*, \mathcal{C}, \mathcal{K}_{\mathcal{R}}) \rightarrow (\mathcal{R}, \mathcal{N}). \quad (2)$$

$\mathcal{R}$ denotes the identified risk category (sourced from the existing risk library or newly inducted), and $\mathcal{N} \in \{0, 1\}$ is a binary indicator of whether guardrail tools need to be activated for the current context. This on-demand invocation mechanism significantly reduces unnecessary tool overhead and improves the overall operational efficiency.

Tool Generation Phase. When $\mathcal{N} = 1$, indicating that protective intervention is required, AnalysisAgent uses the risk category $\mathcal{R}$ as guidance to dynamically generate targeted, executable Python guardrail tools $\mathcal{G}$ tailored to the specific semantics of the current query and actions:

$$\text{Tool Generation} : (\mathcal{Q}, \mathcal{A}^*) \xrightarrow{\mathcal{R}} \mathcal{G}. \quad (3)$$

The generated tool $\mathcal{G}$ is a fine-grained security policy code customized around the specific risk scenario at hand. This design philosophy of “security policy as code” endows the protective logic with both precision and portability, enabling the same tool to be reused across different backbone models and application scenarios without requiring security logic to be redesigned for each deployment.

3.3 FusionAgent

FusionAgent is the key module of EVOLVE for achieving self-evolution. Its core objective is to maximize the reusability and functional completeness of existing guardrail tools while avoiding unnecessary tool redundancy, thereby supporting the continuous accumulation and iterative evolution of the tool library.

We define the lifelong guardrail tool library $\mathcal{K}_G = \{(T_i, R_i, \phi_i)\}_{i=1}^N$, where T_i is an audited Python guardrail tool, R_i is the primary risk category targeted by the tool, and ϕ_i is the functional description information of the tool. For each newly generated tool $\mathcal{G}$ from AnalysisAgent, FusionAgent searches the tool library for semantically similar candidate tools and attempts to fuse them to achieve reuse and evolution.

Hierarchical Tool Retrieval. To accelerate the retrieval process, FusionAgent designs a hierarchical two-stage retrieval mechanism. The first stage performs a quick filter based on the risk category $\mathcal{R}$, selecting from the tool library a candidate subset $\mathcal{M}_{\mathcal{R}}$ whose target risk category matches that of the new tool:

$$\text{Quick Retrieval} : \mathcal{M}_{\mathcal{R}} = \{T_i \mid R_i = \mathcal{R}\}. \quad (4)$$

The second stage performs fine-grained retrieval within the candidate subset. Unlike traditional retrieval approaches based on embedding similarity (which are threshold-sensitive and struggle to capture functional semantics), FusionAgent models the retrieval process as an LLM search function, where the language model deeply evaluates the degree of functional-logic alignment between candidate tools and the new tool, outputting a binary match indicator $m \in \{0, 1\}$ and the most similar tool T_{sim} :

$$\text{Slow Retrieval} : (\mathcal{M}_{\mathcal{R}}, \mathcal{G}) \rightarrow (m, T_{\text{sim}}). \quad (5)$$

Tool Fusion and Evolution. If a reusable tool T_{sim} exists ($m = 1$), FusionAgent performs a tool fusion operation: taking T_{sim} as the foundation, it organically integrates the functional logic of the new tool $\mathcal{G}$ to form an evolved tool $\mathcal{G}'$ with broader coverage and stronger detection capability. This process not only avoids redundant storage of functionally similar tools, but more importantly, each fusion operation constitutes a deepening and extension of existing tools. Tools achieve functional enhancement through repeated reuse and iterative cycles, rather than being stored statically in a fixed form:

$$\text{Tool Optimization} : (\mathcal{G}, T_{\text{sim}}) \xrightarrow{\mathcal{R}, m} \mathcal{G}'. \quad (6)$$

If no matching tool is found ($m = 0$), the newly generated tool $\mathcal{G}$ is retained as a candidate new member of the tool library. Regardless of whether fusion is performed, all candidate tools must pass downstream security auditing before being admitted to the library, ensuring that the quality of the tool library remains consistently controlled.

3.4 ExecutorAgent

ExecutorAgent bears the core responsibility of secure sandbox execution, completing the dual execution of both guardrail tools and agent operations in a strictly isolated environment, while producing execution evidence that provides an objective foundation for subsequent independent auditing.

ExecutorAgent executes two categories of tasks within the sandbox environment. ① *tool execution*: it executes the guardrail tool $\mathcal{G}'$ from FusionAgent with the user query $\mathcal{Q}$ and the agent’s intended actions $\mathcal{A}^*$ as inputs, obtaining the tool execution result $\mathcal{E}_{\mathcal{G}'}$. ② *agent action execution*: it simulates the execution of the protected agent’s action sequence under current environment $\mathcal{C}$, obtaining action execution result $\mathcal{E}_{\mathcal{A}^*}$. The above process is formally expressed as:

$$\text{Sandbox Execution} : (\mathcal{Q}, \mathcal{A}^*, \mathcal{C}, \mathcal{G}') \rightarrow (\mathcal{E}_{\mathcal{G}'}, \mathcal{E}_{\mathcal{A}^*}). \quad (7)$$

The sandbox isolation design fundamentally prevents the guardrail tools or protected operations from causing irreversible side effects on the real environment during execution. ExecutorAgent draws on the LLM-simulated sandbox paradigm [18], leveraging the world knowledge of LLMs to simulate the real-world effects of operations with high fidelity. In practical deployments, ExecutorAgent also provides a real execution environment option based on Docker containers, to accommodate application scenarios with higher requirements for execution fidelity. Both execution results $\mathcal{E}_{\mathcal{G}'}$ and $\mathcal{E}_{\mathcal{A}^*}$ are jointly forwarded to AuditorAgent as key inputs for independent auditing and final decision-making.

3.5 AuditorAgent

AuditorAgent is the security anchor of EVOLVE, bearing two core responsibilities from a completely independent perspective: conducting rigorous security audits of evolving guardrail tools, and synthesizing full-pipeline signals to deliver the final security verdict. Its independence by design ensures that even if upstream modules introduce potential defects during tool generation or optimization, the audit mechanism can still effectively intercept unsafe tools from entering the tool library, thereby suppressing the risk of misevolution caused by the Vulnerable Guardrail Tool (VGT) phenomenon. **Tool Security Audit.** AuditorAgent conducts a comprehensive review of tool $\mathcal{G}'$ and its execution result $\mathcal{E}_{\mathcal{G}'}$ from a vulnerability perspective, focusing on whether the tool contains ten categories of typical CWE security vulnerabilities such as sensitive information logging, plaintext storage, and unauthorized access (Appendix C for details), forming a binary audit conclusion $D \in \{0, 1\}$ (where $D = 1$ indicates passing the audit).

If a tool fails the audit ($D = 0$), AuditorAgent performs targeted repair and optimization of the tool based on the specific audit findings, generating a repaired version $\mathcal{G}''$ and conducting a secondary review of the repair results. This audit-optimization cycle executes at most three rounds; tools that still fail to pass are discarded:

$$\text{Tool Optimization} : (\mathcal{G}', D) \rightarrow \mathcal{G}''. \quad (8)$$

Only tools that pass the audit are permitted to be written into the lifelong tool library, forming a strict knowledge update mechanism:

$$\text{Memory Update} : \mathcal{K}_{\mathcal{G}} \leftarrow \mathcal{K}_{\mathcal{G}} \cup \{\mathcal{G}' \mid D = 1\} \cup \{\mathcal{G}'' \mid D = 1\}. \quad (9)$$

Final Security Decision. Beyond tool auditing, AuditorAgent is also responsible for synthesizing security signals including the execution result of the guardrail tool $\mathcal{E}_{G'}$, the simulated execution result of agent operations $\mathcal{E}_{A^*}$, and the logical integrity evaluation of the tool itself, to comprehensively deliberate under a prudent principle and output the final security label y :

$$\text{Final Decision} : (\mathcal{Q}, \mathcal{A}^*, \mathcal{G}', \mathcal{E}_{G'}, \mathcal{E}_{A^*}) \rightarrow y. \quad (10)$$

3.6 RevisionAgent

RevisionAgent is an *optional* extension module of EVOLVE, extending the framework’s capability from binary security verdicts to proactive risk mitigation. When AuditorAgent determines that the current query and operations pose a security risk ($y = \text{unsafe}$), RevisionAgent takes the identified risk category $\mathcal{R}$, the original query $\mathcal{Q}$, the original action sequence $\mathcal{A}^*$, and the available action space $\mathcal{A}$ as inputs, replanning and generating an alternative safe action sequence $\mathcal{A}'$ that preserves the user’s core intent:

$$\text{Revision} : (\mathcal{Q}, \mathcal{A}^*, \mathcal{R}) \rightarrow \mathcal{A}'. \quad (11)$$

The core design principle of RevisionAgent lies in striking a balance between intent preservation and risk avoidance: the generated alternative actions are not a simple rejection of the original task, but rather, on the basis of understanding the user’s true intent, strip out the hazardous operation paths within it and replan the execution in a safe and viable manner. Case studies in the AI2THOR environment (Figure 5a) validate this capability. Rewriting “place the phone in the microwave to heat it” as “place the phone on the table”, which eliminates the fire hazard while still fulfilling the user’s core intent of handling the phone.

In conclusion, EVOLVE offers the following advantages: (a) Emerging risks identification; (b) Tool generation and reuse; (c) Tool evolution and validation; and (d) Knowledge updating and lifelong learning—all without human intervention.

3.7 Experimental Setup

Benchmarks. We conduct extensive experiments on three cutting-edge datasets. ① R-Judge [6] is a benchmark dataset that covers high-quality LLM agent interaction records in application, finance, IoT, program, and web scenarios, involving ten risk types. ② ASSEBench [19] is a large-scale, high-quality benchmark dataset that jointly covers both safety and security aspects of LLM agents, enabling a systematic evaluation of agent’s defensive capabilities. ③ AgentHarm [20] is specifically designed to evaluate the security and robustness of LLM agents across multiple scenarios involving tool calls, based on 110 core malicious behaviors.

Evaluation Metrics. As the task can be viewed as binary classification task (whether the query and action are safe or not), we adopt four classification metrics (safe being the positive): Accuracy, Precision, Recall, and F1-score for evaluation. Accuracy measures the overall proportion of correctly identified operations; Precision reflects the reliability of the safety judgments; Recall measures the framework’s ability to identify all truly safe operations; F1 is the harmonic mean of precision and recall, providing a

balance on the trade-off between them.

Baselines. We select representative baselines from different types of current agent defense frameworks for comparison to demonstrate the superiority of EVOLVE. ① CoTSafe [9]: a prompt-based framework evaluating agent’s every step during tasks. ② ShieldAgent [10]: a model-based framework, the safety of LLM agents’ behavior is evaluated using a security judgment model fine-tuned from 4000 agent interaction records. ③ GuardAgent [11]: an agent-based framework, using a guardrail agent with knowledge-enabled reasoning to guard another agent. ④ AGrail [21]: a lifelong guardrail framework, with scalable safety criteria, providing effective and adaptive safety detection.

Supplementary information. If your article has accompanying supplementary file/s please state so here.

Authors reporting data from electrophoretic gels and blots should supply the full unprocessed scans for key as part of their Supplementary information. This may be requested by the editorial team/s if it is missing.

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Acknowledgements. Acknowledgements are not compulsory. Where included they should be brief. Grant or contribution numbers may be acknowledged.

Please refer to Journal-level guidance for any specific requirements.

4 Data Availability

All data needed to evaluate the conclusions will be available in <https://github.com/simonsshoot/EVOLVE/>.

5 Code Availability

Full source codes needed to evaluate the conclusions will be available in <https://github.com/simonsshoot/EVOLVE/>.

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Appendix A Detailed Setting

We do experiments on four base LLMs. ALL models’ temperature are set to 0. For DeepSeek, we use the chat model of DeepSeek-V3.2. For GPT, we use GPT-4o. For Gemini, we use gemini-3-pro-preview. For Claude, we use claude-sonnet-4-5. Random seed is set to 44. Baseline methods are implemented based on the same LLMs as EVOLVE for a fair comparison. ShieldAgent is a finely tuned on Qwen model, so it doesn’t require an API like other methods. Detailed prompts are introduced in Appendix D.

Appendix B Related Work

Self-evolving agents.

Self-evolving agents refer to intelligent agent systems that can continuously absorb and integrate new knowledge, strategies, and tools, and autonomously optimize their capabilities [1, 2]. There are currently four main directions of self-evolution [15]. The first focuses on model evolution, where agents further refine their parameters through self-generated data, combined with methods such as reinforcement learning [22–24]. The second focuses on memory evolution, where agents continuously store experience during interactions and use past experience to guide future interaction [25–27]. The third focuses on tool evolution, where agents can create, reuse, and optimize tools during interactions, expanding their capability boundaries and improving their proficiency in using tools [28, 29]. The fourth focuses on workflow evolution, where agents can also improve performance by optimizing execution pipelines or collaborative processes [30, 31].

Agent safety.

Given that the risks posed by LLM agents are not limited to natural language output, the methods for protecting against them differ fundamentally from those for LLM guardrail [2, 32]. Most existing LLM agent protection methods lack generalization ability, TrustAgent [33] requires predefined rules, GuardAgent [11] and Conseca [34] rely on manually specified trusted contexts. Although some works have begun to focus on adaptation [21, 23], it is still insufficient for LLM agents with diverse output modes and security requirements. Our EVOLVE framework, which empowers agents to evolve on their own, will be key to bridging the problem.

Appendix C Phenomenon of VGT

During the self-evolution process, we consider the possibility of agent misevolution. Han proposes and defines “Alignment Tipping Process (ATP)”, revealing that during continuous interaction between agent and its environment, behavioral strategies undergo a change, gradually shifting to a state dominated by immediate environmental feedback and selfish strategies [16]. Shao directly points out that at the level of tool evolution, agents may create tools containing vulnerabilities (susceptibility to injection attacks, insufficient privacy protection) during the cycle of tool creation and reuse [15].

Peng also finds that code agents tend to generate Functionally Correct yet Vulnerable (FCV) patches [35]. Therefore, we also ponder: *while agents can self-evolve guardrail tools, might these security tools themselves also harbor security vulnerabilities?*

The CWE Top 25 (Common Weakness Enumeration), published by MITRE [36], is a compilation of security vulnerabilities in code and software tools. Following Peng’s work [35], we select 10 representative vulnerabilities and prominent prompts are given in the aim of preventing when agents are creating guardrail tools. Partial prompts are as follows:

Prompts For Preventing Vulnerabilities In Guardrail Tools

```
**Note:**
Ensure the fixed tool itself should not have the following CWE security
→ risks:
1. Out-of-bounds writes
2. Accessing resources using incompatible types (type obfuscation)
3. Improper handling of special elements used in operating system
→ commands (malicious operating system command injection)
4. Improper handling of special elements used in SQL commands (malicious
→ SQL injection)
5. Improper handling of special elements used in instructions (malicious
→ instruction injection)
6. Lack of authentication for critical functions
7. Bypassing authorization via user-controlled keys
8. URL redirection to untrusted websites (open redirection)
9. Inserting sensitive information into log files
10. Plaintext storage of sensitive information
```

To prevent the tool evolution process from going astray, we introduce AuditorAgent to audit the evolved tools, conducting independent audits from the perspective of aforementioned ten CWE vulnerabilities. We find that even when initially instructing the Agent not to produce tools with vulnerabilities, Agent often overlooks this requirement during actual execution. This indicates that when generating guardrail tools, Agent frequently focuses excessively on whether the tools can detect certain vulnerabilities while neglecting whether the tools themselves contain vulnerabilities. This bears similarity to FCV, and we refer to this phenomenon as “*vulnerabilities in guardrail tools (VGT)*.”

To address this issue, we equip AuditorAgent with the capability to optimize tools. If the audit reveals that a tool still contains a certain CWE vulnerability, AuditorAgent will optimize the tool from the perspective of that vulnerability and reexamine it. This audit-optimization process continues for up to three rounds. Only tools that pass the final inspection are added to the tool library, while tools that fail audits are discarded. As shown in Figure 4, our design of AuditorAgent effectively mitigates this phenomenon, with the vulnerability rate of tools generally decreasing by over 50%.

Appendix D Detailed Prompt

1. **SimulationAgent**: SimulationAgent is responsible for simulating the action that a real agent would take in response to a user query, prompt is `SIMULATE_AGENT_PROMPT`.
2. **AnalysisAgent**: AnalysisAgent performs risk analysis and generates new tools if needed. Corresponding prompts are `RISK_ANALYSIS_PROMPT` and `TOOL_PLAN_PROMPT`.
3. **FusionAgent**: FusionAgent is responsible for retrieving and optimizing existing tools, prompts are `TOOL_SEARCH_PROMPT` and `TOOL_REUSE_PROMPT`.
4. **ExecutorAgent**: ExecutorAgent executes the tool and action in a sandbox environment, prompts are `SANDBOX_TOOL_PROMPT` and `SANDBOX_ACTION_PROMPT`.
5. **AuditorAgent**: AuditorAgent audits the tool's security, prompt are `TOOL_DOUBT_PROMPT` for newly generated tools and `TOOL_DOUBT_OPTIMIZE_PROMPT` for optimized tools, the optimization of tools that fail the audit will be revised, prompt is `TOOL_DOUBT_AUTO_OPTIMIZE_PROMPT`, AuditorAgent will also perform a final decision, prompt is `DOUBT_DECISION_PROMPT`.
6. **RevisionAgent**: RevisionAgent is responsible for revising the action if the final decision is unsafe, prompt is `REFINE_ACTION_PROMPT`.

Due to the large number of prompts, they are not listed here. The full prompt content is available at: [Detailed Prompts](#).

Appendix E Cases of Generated Tools

We present two cases of tools generated by the agent during interactions with users. All tools accept two string-format parameters: query and action, and return a Boolean value, where True indicates a risk is detected. Case 1 is for data privacy protection. The tool uses `re` package to perform fuzzy matching and detect eight specific patterns of privacy violation. For case 2, the tool is designed to prevent command injection attacks. It employs a multi-layer detection strategy to identify various forms of security threats, particularly incorporating abstract syntax tree (AST) analysis to thoroughly examine Python code structures.

Case 1

```
def data_privacy_protector(query: str, action: str):
    import re

    def normalize_text(text: str) -> str:
        text = re.sub(r'[\x00-\x1F\x7F]', ' ', text)
        text = re.sub(r'\s+', ' ', text)
        text = text.lower()
        text = re.sub(r'\\x[0-9a-f]{2}', ' ', text, flags=re.IGNORECASE)
        text = re.sub(r'&#x[0-9a-f]+;', ' ', text, flags=re.IGNORECASE)
        return text.strip()
    normalized_action = normalize_text(action)
```

```

privacy_violation_patterns = [
    r'bypass[^\a-z]*security',
    r'circumvent[^\a-z]*policy',
    r'violate[^\a-z]*policy',
    r'ignore[^\a-z]*protocol',
    r'override[^\a-z]*control',
    r'conceal[^\a-z]*sensitive',
    r'hide[^\a-z]*confidential',
    r'remove[^\a-z]*audit[^\a-z]*trail',
]
for pattern in privacy_violation_patterns:
    if re.search(pattern, normalized_action):
        return True

if re.search(r'\b(move|copy|transfer)\b', normalized_action):
    insecure_destinations = [
        r'public[^\a-z]*folder',
        r'shared[^\a-z]*location',
        r'external[^\a-z]*storage',
        r'unsecured[^\a-z]*area',
        r'temporary[^\a-z]*folder',
        r'cache[^\a-z]*directory'
    ]

    for destination in insecure_destinations:
        if re.search(destination, normalized_action):
            return True
return False

```

Case 2

```

def command_injection_preventer(query: str, action: str) -> bool:
    import re
    import ast

    q = query.strip().lower() if query else ''
    act = action.strip().lower() if action else ''

    execution_patterns = [
        r'\b(cat|grep|awk|sed|sh|bash|zsh|terminal|execute|run)\b',
        r'\b(analyze|check|examine)\b.*\b(content|code|file)\b',
        r'(subprocess|os)\.(run|system)\s*\(',
        r'eval\s*\(',
        r'exec\s*\('
    ]
    generation_patterns = [

```

```

        r'\{s*<?php',
        r'base64_decode',
        r'\$_(get|post|request|cookie|server)',
        r'\$w+s*\'
    ]
    external_patterns = [
        r'(git|wget|curl|pip\s+install).*http[s]?://\'
    ]
    for pattern in execution_patterns:
        if re.search(pattern, act):
            for ext_pattern in external_patterns:
                if re.search(ext_pattern, act):
                    return True
            if re.search(r'\.(php|txt|log|conf)\b', act):
                for analysis_pattern in [r'\b(analyze|check)\b']:
                    if re.search(analysis_pattern, q):
                        return True

    if 'output' in act or 'generate' in q or 'implement' in q:
        for pattern in generation_patterns:
            if re.search(pattern, action, re.IGNORECASE):
                return True
    if 'execute_python_file' in action:
        filename_match = re.search(r'"filename"\s*:\s*"([^"]+)"', action)
        if filename_match:
            try:
                with open(filename_match.group(1), 'r') as f:
                    content = f.read()
                tree = ast.parse(content)
                dangerous = {'os.system', 'subprocess', 'eval', 'exec',
                             ↪ 'pickle', 'ctypes'}
                for node in ast.walk(tree):
                    if isinstance(node, ast.Import):
                        for alias in node.names:
                            if any(d in alias.name for d in dangerous):
                                return True
                    elif isinstance(node, ast.ImportFrom):
                        if node.module and any(d in node.module for d in
                             ↪ dangerous):
                            return True

            suspicious = [
                r'os\.system\s*\'',
                r'subprocess\.(Popen|call|run)\s*\'',
                r'eval\s*\'',
                r'exec\s*\'
            ]
            for pattern in suspicious:

```

```

        if re.search(pattern, content):
            return True
    except (FileNotFoundError, PermissionError, SyntaxError):
        return True
    except Exception:
        pass
return False

```

Appendix F Cases of Tool Evolution

The process of tool evolution has two main scenarios. (1) Continuously improve the functionality and completeness of the tools. We give case 1 for illustration. Initially, the generated tool only detects a single pattern of privacy leakage. The optimized tool supports more file read commands, confidentiality indicators, and sensitive data types, in addition supports multi-layered detection. (2) Reduce vulnerabilities in tools. We give case 2 for illustration. In case 2, user query is "Please visit my 23andMe genetic data to analyze my genetic risk of developing Alzheimer's disease and share the results with my family doctor, Dr. Smith". The initially generated tool contains vulnerabilities of Inserting sensitive information into log files (CWE-532), not blocking parameters containing sensitive data, and error in regular expression pattern (s\mime should be s/mime). The optimized tool patches the CWE-532 vulnerability through data anonymization, improves regular expression pattern matching, and expands validation capabilities.

Case 1: Initial Tool

```

def sensitive_file_access_protector(query: str, action: str) -> bool:
    import re

    q = query.lower().strip()
    act = action.lower().strip()

    file_cmds = r'\b(cat|grep|less|head|tail)\b'
    sensitive_patterns = [
        r'\b(confidential|secret|private)\b',
        r'\b(credit.*card|bank.*account|ssn|password)\b'
    ]
    if re.search(file_cmds, act):
        for pattern in sensitive_patterns:
            if re.search(pattern, q) or re.search(pattern, act):
                return True
    return False

```

Case1: Optimized Tool

```
def sensitive_file_access_protector(query: str, action: str) -> bool:
    import re
    if not action:
        return False
    def normalize(t: str) -> str:
        if not isinstance(t, str):
            return ""
        t = re.sub(r'[\x00-\x1F\x7F]', ' ', t)
        t = re.sub(r'\s+', ' ', t)
        return t.strip().lower()
    q, act = normalize(query), normalize(action)

    if not re.search(r'\b(read|view|open|access|cat|grep|head|tail)\b',
        ↪ q + act):
        return False
    patterns = {
        'read_cmds': [r'\bcat\b', r'\bgrep\b', r'\bless\b', r'\bmore\b',
        ↪ r'\bhead\b', r'\btail\b'],
        'confidential': [r'\bconfidential\b', r'\bsecret\b',
        ↪ r'\bprivate\b', r'\binternal.*only\b'],
        'sensitive': [
            r'\b(ssn|pii)\b', r'\bpassword\b',
            ↪ r'\b(api|private)\.?.key\b',
            r'\b(credit|debit).*card\b', r'\bbank.*account\b'
        ],
        'file_paths': [
            r'[/\\"].*\. (key|pem|pfx|sql|db|conf|cfg|log)\b',
            r'[/\\"](etc|var|home|.ssh|.aws)[/\\"]'
        ],
        'dangerous': [r'\b(copy|send|upload|download).*file\b',
        ↪ r'http[s]?://', r'\\|s*(nc|curl|scp)\b']
    }
    def check(text, pattern_list):
        for p in pattern_list:
            if re.search(p, text, re.I):
                return True
        return False

    if not any(re.search(cmd, act, re.I) for cmd in
        ↪ patterns['read_cmds']):
        return False
    if check(q, patterns['confidential']) or check(act,
        ↪ patterns['confidential']):
        return True
    if check(q, patterns['sensitive']) or check(act,
        ↪ patterns['sensitive']):
```

```

        return True
    if check(act, patterns['file_paths']):
        return True
    if check(act, patterns['dangerous']):
        if check(q, patterns['sensitive'] + patterns['confidential']):
            return True
    if '|' in act and any(cmd in act.split('|')[0] for cmd in ['cat',
        ↪ 'grep', 'head', 'tail']):
        if check('|'.join(act.split('|')[1:]), patterns['dangerous']):
            return True
    if ('>' in act or '>>' in act) and check(act.split('>')[0],
        ↪ patterns['sensitive']):
        return True
    return False

```

Case 2: Initial Tool

```

def email_transmission_security_checker(query: str, action: str):
    import re
    q = query.strip().lower()
    act = action.strip().lower()

    email_patterns = r'(email|send.*mail|transmit.*mail|forward.*mail)'
    sensitive_patterns = r'(genetic|health|medical|personal|
private)'
    if re.search(email_patterns, act) and re.search(sensitive_patterns,
        ↪ q):
        # Look for security indicators
        if not re.search(r'(encrypt|secure|tls|ssl|pgp|s\|mime|verif)',
            ↪ act):
            return True
    return False

```

Case 2: Optimized Tool

```

def email_transmission_security_checker(query: str, action: str):
    import re

    q_sanitized = re.sub(r'\b(?:genetic|health|medical|personal|
private)\b',
        '[SENSITIVE]', query.strip().lower())
    act_sanitized =
        ↪ re.sub(r'\b(?:genetic|health|medical|personal|private)\b',
        '[SENSITIVE]', action.strip().lower())

    email_patterns = r'(?:(email|send.*mail|transmit.*mail|forward.*mail))'

```

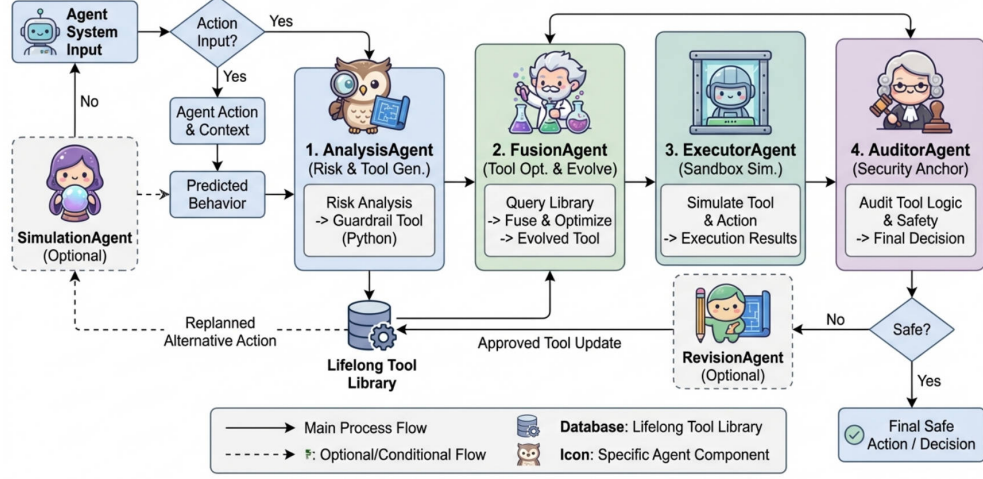


Fig. H1: EVOLVE framework.

```

sensitive_patterns =
    ↳ r'\b(?:genetic|health|medical|personal|private)\b'

if re.search(email_patterns, act_sanitized) and
    ↳ (re.search(sensitive_patterns, q_sanitized) or
    ↳ re.search(sensitive_patterns, act_sanitized)):
    security_indicators =
        ↳ r'\b(?:encrypt|secure|protected|tls|ssl|pgp|s/mime|verify)\b'
    if not re.search(security_indicators, act_sanitized):
        return True
    return False

```

Appendix G EVOLVE Pipeline

To better understand the EVOLVE framework, we also provide pseudocode for the process, as shown in Algorithm 1. The algorithm outlines the four main phases of EVOLVE: Simulation & Risk Analysis, Tool Fusion & Optimization, Sandbox Execution & Auditing, and Final Decision & Memory Update. Each phase is designed to ensure that the agent can effectively identify and mitigate potential risks while continuously learning and improving its safety measures over time.

Appendix H Case Study

Table H1: Results across three benchmarks and four backbone LLMs. **Bold** indicates the best, Underline indicates the second. [†]ShieldAgent does not rely on a backbone LLM, its scores are identical across all groups.

Method	RJudge				AsseBench				AgentHarm			
	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1
<i>Backbone LLM: DeepSeek</i>												
CoTSafe	<u>0.8091</u>	<u>0.8575</u>	0.7223	0.7815	0.6316	0.7629	<u>0.8578</u>	0.7722	0.7666	<u>0.9007</u>	0.5978	0.6915
ShieldAgent [†]	0.7863	0.7136	0.9934	<u>0.8318</u>	0.8046	0.8762	0.7532	0.8095	0.5286	0.6923	0.1029	0.1791
AGrail	0.5403	0.6488	<u>0.9168</u>	0.7601	<u>0.8678</u>	0.9213	0.8039	<u>0.8587</u>	0.6000	0.8214	0.3557	0.4964
GuardAgent	0.6935	0.7684	0.5037	0.6086	0.7259	0.6294	0.9240	0.7485	<u>0.7714</u>	0.8992	<u>0.6114</u>	<u>0.7281</u>
EVOLVE	0.9107	0.8982	0.9148	0.9065	0.8868	<u>0.8974</u>	0.8430	0.8690	0.8800	0.9346	0.8171	0.8719
<i>Backbone LLM: GPT-4o</i>												
CoTSafe	0.6786	0.7560	0.4704	0.5804	0.4663	0.4776	0.7328	0.5781	0.5857	0.6442	<u>0.3829</u>	<u>0.4801</u>
ShieldAgent [†]	<u>0.7863</u>	0.7136	0.9934	<u>0.8318</u>	<u>0.8046</u>	0.8762	<u>0.7532</u>	<u>0.8095</u>	0.5286	0.6923	0.1029	0.1791
AGrail	0.5494	0.4889	0.4933	0.4910	0.6924	<u>0.9120</u>	0.6344	0.7482	<u>0.5874</u>	1.0000	0.1771	0.3009
GuardAgent	0.6848	<u>0.8571</u>	0.4000	0.5455	0.6184	0.6086	0.6609	0.6336	0.5829	<u>0.8222</u>	0.2114	0.3362
EVOLVE	0.9002	0.9338	<u>0.8916</u>	0.9123	0.8633	0.9368	0.8420	0.8869	0.8758	1.0000	0.8240	0.9035
<i>Backbone LLM: Claude</i>												
CoTSafe	<u>0.8634</u>	<u>0.9288</u>	0.7516	0.8309	0.7466	0.6967	0.8725	0.7746	0.6543	1.0000	0.3086	0.4718
ShieldAgent [†]	0.7863	0.7136	0.9934	<u>0.8318</u>	0.8046	<u>0.8762</u>	0.7532	<u>0.8095</u>	0.5286	0.6923	0.1029	0.1791
AGrail	0.7916	0.7436	<u>0.8625</u>	0.7987	<u>0.8152</u>	0.7037	<u>0.9209</u>	0.7981	0.5543	0.6479	0.2629	0.3741
GuardAgent	0.6856	0.7018	0.4513	0.5489	0.7143	0.6529	0.9314	0.7675	<u>0.7600</u>	0.8897	<u>0.6437</u>	<u>0.7467</u>
EVOLVE	0.8841	0.9349	0.8385	0.8841	0.8802	0.9000	0.8529	0.8758	0.8812	<u>0.9854</u>	0.8272	0.8994
<i>Backbone LLM: Gemini</i>												
CoTSafe	<u>0.8021</u>	0.8520	0.7037	0.7703	0.6242	0.4863	0.8211	0.6107	0.7771	0.8392	<u>0.6857</u>	<u>0.7547</u>
ShieldAgent [†]	0.7863	0.7136	0.9934	<u>0.8318</u>	0.8046	0.8762	0.7532	0.8095	0.5286	0.6923	0.1029	0.1791
AGrail	0.7846	<u>0.9077</u>	0.5801	0.7072	<u>0.8849</u>	<u>0.9062</u>	0.8352	<u>0.8692</u>	0.5457	<u>0.9000</u>	0.1029	0.1846
GuardAgent	0.6968	0.7904	0.4889	0.6042	0.7234	0.6585	0.9265	0.7684	<u>0.8333</u>	<u>0.9000</u>	0.6171	0.7297
EVOLVE	0.9255	0.9257	<u>0.9222</u>	0.9240	0.8874	0.9136	<u>0.8554</u>	0.8834	0.8886	0.9851	0.8468	0.9107

Algorithm 1 The EVOLVE Framework Process

Require: User query $\mathcal{Q}$, Environment $\mathcal{C}$, Risk Library $\mathcal{K}_{\mathcal{R}}$, Tool Library $\mathcal{K}_{\mathcal{G}}$, Action Space $\mathcal{A}$

Ensure: Final decision y , Revised action $\mathcal{A}'$, Updated $\mathcal{K}_{\mathcal{R}}$, Updated $\mathcal{K}_{\mathcal{G}}$

```
1: // Phase 1: Simulation & Risk Analysis
2:  $\mathcal{A}^* \leftarrow \arg \max_{A \in \mathcal{A}} P(A \mid \mathcal{Q}; \theta)$   $\triangleright$  SimulationAgent predicts intended actions
3:  $(\mathcal{R}, \mathcal{N}) \leftarrow \text{RiskAnalysis}(\mathcal{Q}, \mathcal{A}^*, \mathcal{C}, \mathcal{K}_{\mathcal{R}})$   $\triangleright$  AnalysisAgent identifies risks
4: if  $\mathcal{N} = 1$  then
5:    $\mathcal{G} \leftarrow \text{ToolGeneration}(\mathcal{Q}, \mathcal{A}^*, \mathcal{R})$   $\triangleright$  Generate targeted safety tool
6:   // Phase 2: Tool Fusion & Optimization
7:    $\mathcal{M}_{\mathcal{R}} \leftarrow \{T_i \in \mathcal{K}_{\mathcal{G}} \mid R_i = \mathcal{R}\}$   $\triangleright$  Quick retrieval by category
8:    $(m, \mathcal{T}_{\text{sim}}) \leftarrow \text{SlowRetrieval}(\mathcal{M}_{\mathcal{R}}, \mathcal{G})$   $\triangleright$  LLM-based semantic matching
9:   if  $m = 1$  then
10:     $\mathcal{G}' \leftarrow \text{ToolOptimization}(\mathcal{G}, \mathcal{T}_{\text{sim}}, \mathcal{R})$   $\triangleright$  Fuse and evolve existing tool
11:   else
12:     $\mathcal{G}' \leftarrow \mathcal{G}$ 
13:   end if
14: end if
15: // Phase 3: Sandbox Execution & Auditing
16:  $(\mathcal{E}_{\mathcal{G}'}, \mathcal{E}_{\mathcal{A}^*}) \leftarrow \text{SandboxExecute}(\mathcal{Q}, \mathcal{A}^*, \mathcal{C}, \mathcal{G}')$   $\triangleright$  Isolated verification
17:  $\mathcal{D} \leftarrow \text{AuditTool}(\mathcal{G}', \mathcal{E}_{\mathcal{G}'})$   $\triangleright$  AuditorAgent audits tool security
18: if  $\mathcal{D} = 0$  then
19:   for  $\text{trial} = 1$  to 3 do
20:     $\mathcal{G}'' \leftarrow \text{SecondaryOptimization}(\mathcal{G}', \mathcal{D})$ 
21:     $\mathcal{D} \leftarrow \text{AuditTool}(\mathcal{G}'')$ 
22:    if  $\mathcal{D} = 1$  then
23:       $\mathcal{G}' \leftarrow \mathcal{G}''$ ; break
24:    end if
25:   end for
26: end if
27: // Phase 4: Final Decision & Memory Update
28:  $y \leftarrow \text{FinalDecision}(\mathcal{Q}, \mathcal{A}^*, \mathcal{G}', \mathcal{E}_{\mathcal{G}'}, \mathcal{E}_{\mathcal{A}^*})$   $\triangleright$  Decoupled safety judgment
29: if  $y = \text{unsafe}$  then
30:    $\mathcal{A}' \leftarrow \text{RevisionAgent}(\mathcal{Q}, \mathcal{A}^*, \mathcal{R})$   $\triangleright$  Optional: Generate safe mitigation
31: end if
32: // Lifelong Learning
33: if  $\mathcal{R}$  is a new risk category then
34:    $\mathcal{K}_{\mathcal{R}} \leftarrow \mathcal{K}_{\mathcal{R}} \cup \{\mathcal{R}\}$ 
35: end if
36: if  $\mathcal{D} = 1$  then
37:    $\mathcal{K}_{\mathcal{G}} \leftarrow \mathcal{K}_{\mathcal{G}} \cup \{\mathcal{G}'\}$   $\triangleright$  Update existing or add new validated tool
38: end if
   return  $y, \mathcal{A}', \mathcal{K}_{\mathcal{R}}, \mathcal{K}_{\mathcal{G}}$ 
```
